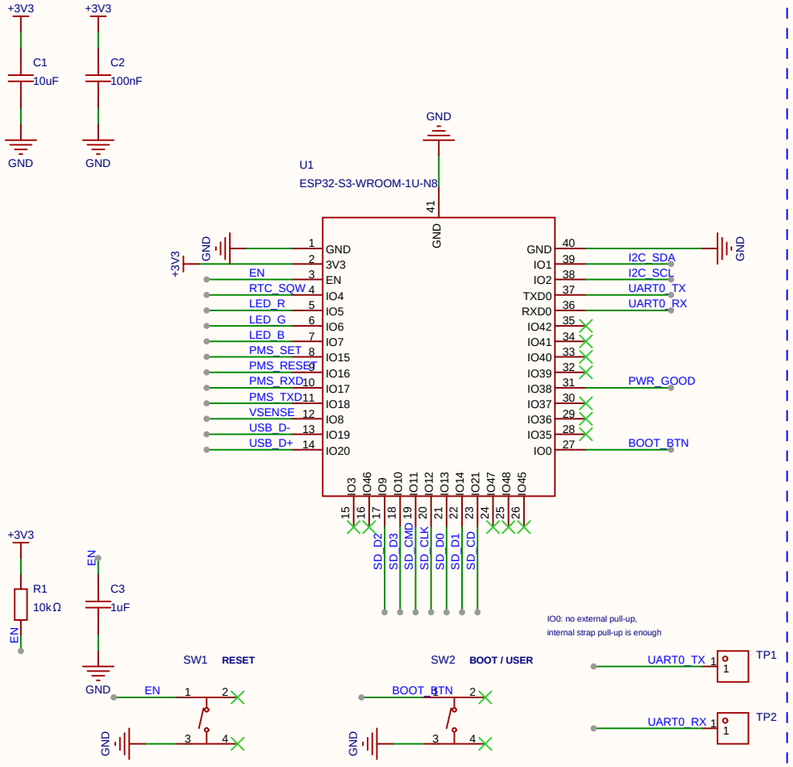
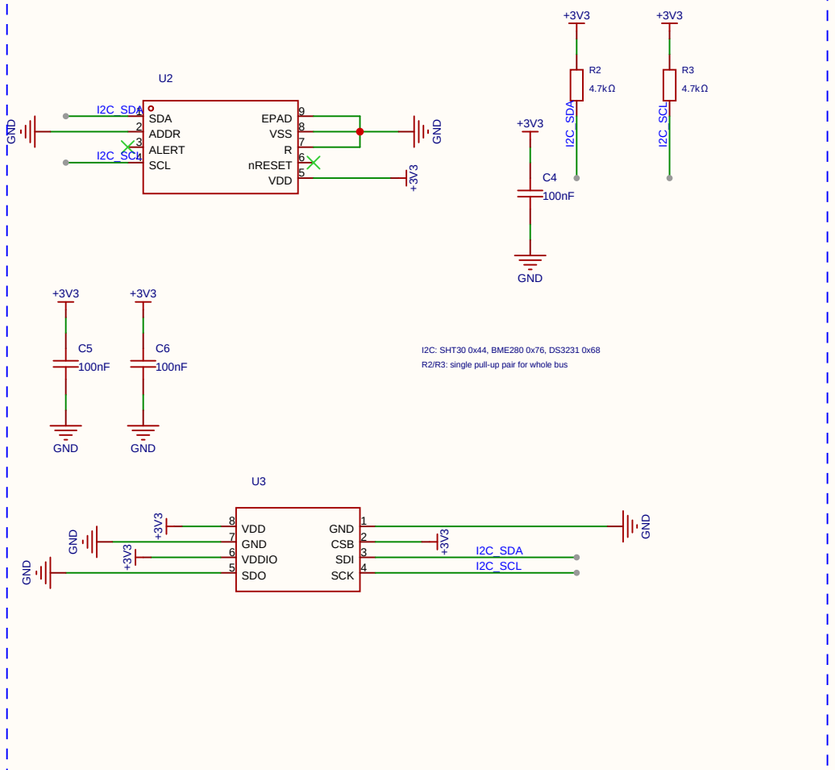
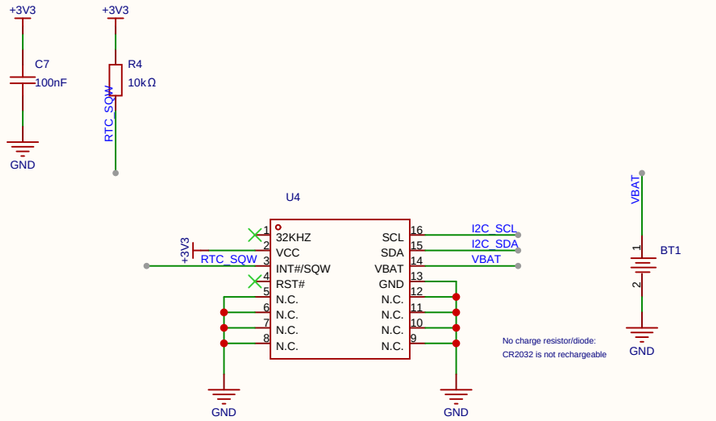
A. MCU core (U1) + B. Buttons (SW1, SW2)



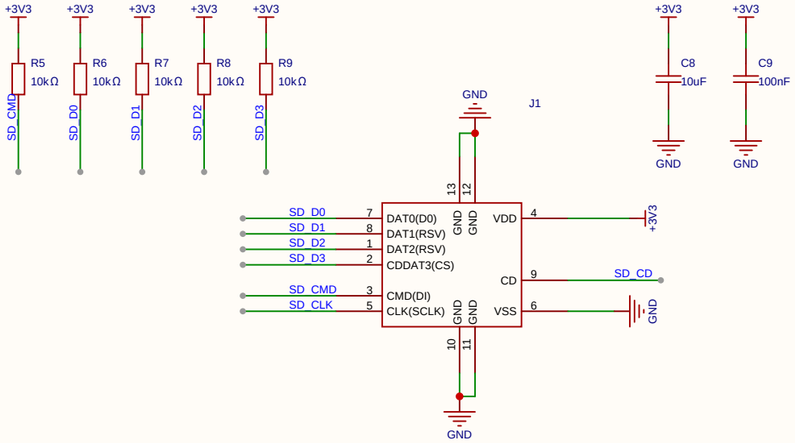
C. I2C sensors (U2 SHT30, U3 BME280)



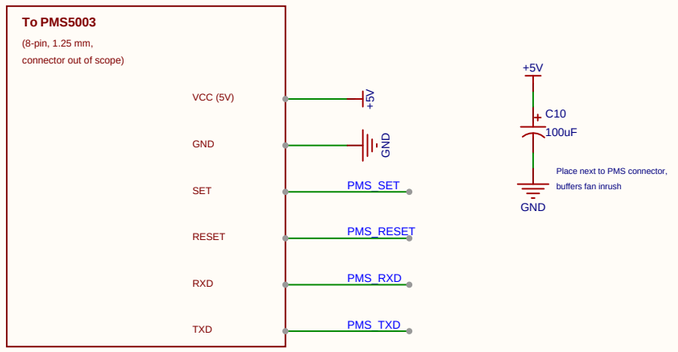
D. RTC (U4 DS3231)



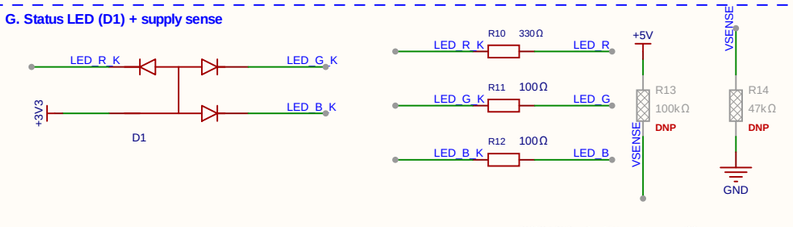
E. microSD (J1)



F. PMS5003 interface (off-board)



G. Status LED (D1) + supply sense



H. Notes

NOTES

1. Power distribution, connectors and OC/OV protection are out of scope (per task).
   Rails shown as +5V / +3V3 ports.
2. I2C addresses: SHT30 0x44, BME280 0x76, DS3231 0x68.
   Single pull-up pair R2/R3 for the whole bus.
3. IO0 doubles as BOOT strap and user button (no external pull-up).
   Strapping pins IO3/45/46 unused.
4. IO33-IO37 left free so an N8R8 (octal PSRAM) module can be fitted
   without layout change.
5. DS3231 backup cell has no charge path (CR2032 is not rechargeable).
6. Component count: 4 ICs/modules, 1 socket, 1 battery holder, 1 LED,
   2 switches, 12 resistors, 10 capacitors, 2 test pads = 33 parts (+2 DNP).
7. Substitutions: J1 = TF-01A (named CD pin) instead of Molex 1040310811;
   D1 = TZ1615RGB3U (1615 RGB, common anode); C10 = 100uF 16V alu-elec.
8. U2 nRESET left open (internal pull-up, per SHT3x datasheet).

Schematic	Praan AQ Node - MCU and Peripherals	Create at	2026-10-11
		Update at	2026-10-11
Board	praan-ay	Page	MCU_Peripherals
Drawn	Pranay	Praan AQ Node - Controller	
Reviewed	Pranay		
		Version	Size / Page 1 Total 1
EasyEDA		A	A3 / Praan